Problem Statement: Customer Offer Engagement Dashboard

Objective

Build an interactive Power BI dashboard using the Customer, Events, Offers, and Offers Metadata datasets to enable marketing teams and analysts to:

- Understand customer engagement with promotional offers (received, viewed, completed)
- Calculate offer redemption rates and analyze transaction behavior post-offer
- Segment customers by demographics (age, gender, income) and engagement levels
- Identify most effective offer types, channels, and reward strategies
- Enable ad hoc analysis of coverage, conversion, and spend metrics to inform marketing strategies

Key Requirements

Data Ingestion & Cleaning

- Connect Power BI to the CSV files: customers.csv, events.csv, offers.csv, and data_dictionary.csv.
- Parse JSON strings in the events.value field to extract offer_id or transaction amount.
- Standardize customer demographic fields: convert membership date to date type, handle missing gender and income values.
- Validate offer metadata fields: ensure offer_id matches between offers.csv and events.csv.

Data Modeling

- Construct a star schema with a central “CustomerEvents” table linking Customer and Offer dimensions.
- Create lookup tables: Customer (customer_id, demographics), Offer (offer_id, type, difficulty, reward, duration, channels).
- Create a fact table by unpivoting events by event type, storing timestamps and transaction amounts.
- Derive measures: count of offers sent/viewed/completed per customer and per offer.

Segmentation & Analysis

- Segment customers into cohorts based on demographics and engagement (e.g., high vs low engagement).
- Compute redemption rates for each offer type and channel.
- Analyze average transaction amounts and frequency before and after offer completion.

Visualization & User Experience

- Page 1: Overview – KPI cards: Total Customers, Offers Sent, Offers Completed, Overall Redemption Rate, Average Transaction Amount.
- Page 2: Offer Funnel – Funnel chart showing volumes: offer received → offer viewed → offer completed → transaction.

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- Page 3: Customer Segmentation – Slicer-driven charts: demographics vs redemption rate, heatmap of redemption by age/income.
- Page 4: Offer Performance – Bar charts: redemption rate by offer type, reward, channel; scatter plot of difficulty vs completion rate.
- Page 5: Transaction Behavior – Line chart: average spend over time for customers who completed offers vs those who did not.

Testing & Optimization

- Validate DAX measures against manual aggregations in Power Query for a sample of 100 customers.
- Cross-check extracted JSON values for correctness (offer_id, transaction amount).
- Optimize data model: remove unused columns, reduce cardinality on text fields (e.g., channels).
- Ensure dashboard refresh time is under 3 minutes for the full dataset (~17,000 customers, ~100,000 events).

Documentation & Deployment

- User Guide: One-page PDF explaining how to refresh data, navigate pages, and use slicers for segmentation.
- Technical Document: Data model ERD, list of DAX formulas, Power Query M steps for JSON extraction, instructions for adding new offers.
- Publish to Power BI Service: Upload .pbix, configure workspace permissions, schedule daily or weekly data refresh.

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